

event detection. Also, it is not sensitive to outliers like amplitude envelopes. To calculate RMS energy within a frame, we first square the amplitudes within the frame and then sum them up. We then divide this figure by the total number of samples in the frame and take the square root, which gives us the RMS energy of that specific frame.

```
librosa.feature.rms(signal, frame_
length, hop_length)
```

Zero crossing rate (ZCR)

The ZCR studies the rate at which a signal's amplitude changes within a frame. It can be used for identifying the percussion sound in music, as signals of that type of sound often have a fluctuating crossing rate. However, this feature is often used in human speech detection. To calculate the ZCR in a specific frame, we apply the sigmoid function to two continuous samples and divide their summation by 2.

```
librosa.feature.zero_crossing_
rate(signal, frame_length, hop_length)
```

Fourier transform and spectrogram

The features described above do not provide any information on the frequency of the given signal. To get that information, we apply the discrete Fourier transform on the audio signal, which provides us with the frequency and magnitude of each frequency component in the audio signal. The code for plotting the magnitude spectrum of our audio signal is given below:

```
def plot_magnitude_spectrum(signal, sr,
title, f_ratio=1):
    x = np.fft.fft(signal)
    x_mag = np.absolute(x)
    plt.figure(figsize=(18, 5))
    f = np.linspace(0, sr, len(x_mag))
    f_bins = int(len(x_mag)*f_ratio)
    plt.plot(f[:f_bins], x_mag[:f_bins])
    plt.xlabel('Frequency (Hz)')
    plt.title(title)
```

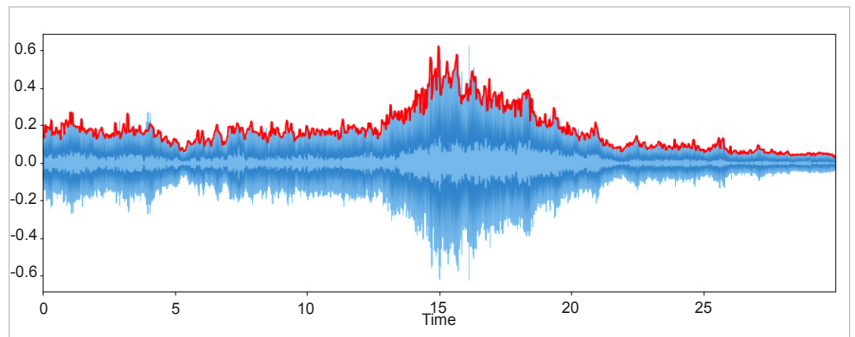


Figure 4: Amplitude envelope of the *wav* file

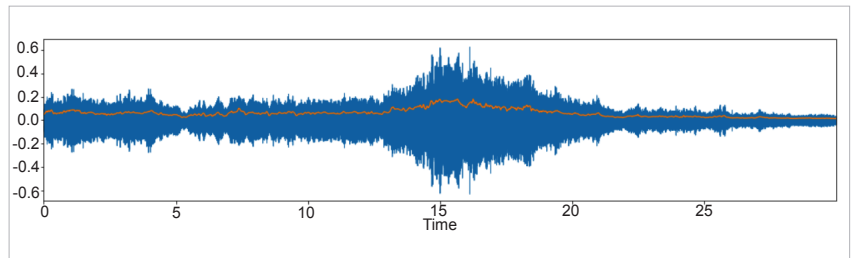


Figure 5: RMS energies of the signal

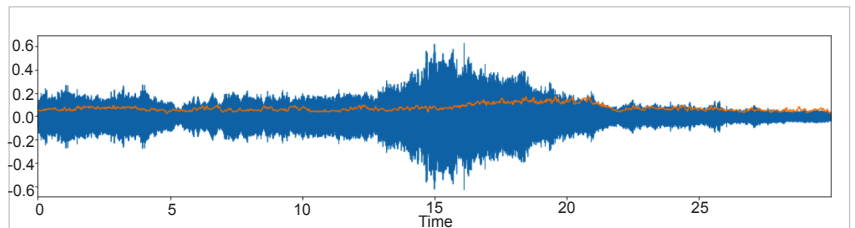


Figure 6: ZCR of an audio signal

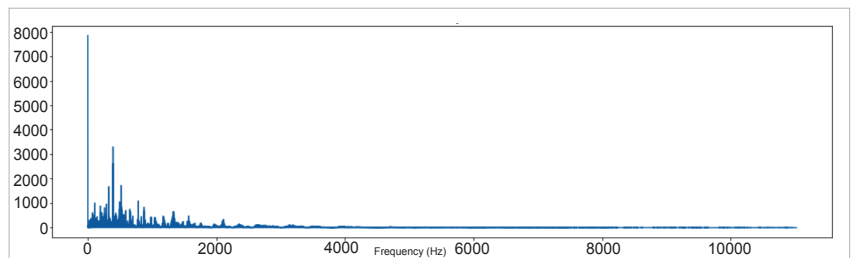


Figure 7: Magnitude spectrum of a signal

Spectrograms using Librosa

The frequency spectrum (Figure 7) shows the frequency domain. However, to obtain more details of the signal, we employ time-series analysis in the form of spectrograms. A spectrogram is a visual representation of the spectrum of frequencies of a signal as it varies with time. Additionally, it captures amplitude variations in the signal. Spectrograms

are widely used in the fields of music, speech processing and linguistics. They can be used to identify words phonetically as well as the sounds of various calls of animals. Spectrograms can be obtained by applying the short time Fourier transform (STFT). In STFT, we apply discrete Fourier transform to each specific frame and show the results accordingly.